

Problem Statement

DevArchitect AI – Multi-Agent Software Engineering Assistant

Background

Modern software development involves large, complex codebases with multiple programming languages, frameworks, and dependencies. As projects grow, developers face challenges in understanding repository architecture, maintaining code quality, identifying security vulnerabilities, generating documentation, and onboarding new team members. Existing development tools often address these tasks independently, forcing developers to switch between multiple platforms and reducing overall productivity.

Problem

There is no unified intelligent platform that can comprehensively analyze a software repository, understand its architecture, review code quality, detect security issues, generate documentation, and answer repository-specific questions using contextual knowledge. This fragmented approach increases development time, maintenance effort, and the likelihood of software defects.

Proposed Solution

DevArchitect AI is a Multi-Agent Software Engineering Assistant powered by IBM watsonx.ai and Granite Models. It combines specialized AI agents for repository analysis, architecture understanding, code review, security assessment, documentation generation, and Retrieval-Augmented Generation (RAG)-based repository Q&A. The platform enables developers to upload a GitHub repository or project ZIP and receive comprehensive insights through an interactive dashboard, improving productivity, software quality, and development efficiency.

Objectives

- Analyze repository structure and technology stack.
- Generate architecture summaries and dependency diagrams.
- Perform intelligent code reviews and detect code quality issues.
- Identify security vulnerabilities and recommend remediation.
- Automatically generate README files and technical documentation.
- Enable natural language interaction with the codebase using RAG.
- Improve developer productivity through AI-driven software engineering assistance.

Expected Outcome

The solution provides a unified, enterprise-ready AI platform that simplifies repository understanding, enhances code quality, strengthens software security, automates documentation, and accelerates the software development lifecycle through intelligent multi-agent collaboration.